CSIT228 | OBJECT-ORIENTED PROGRAMMING 2

Project Documentation

Initial documentation: Parts 1 to 3. The later parts are filled in as the project grows.

Project title	Item Borrowing System
Group name	Tulo ka TAO
Section	G02
Date	10/29/2026

Group members

Name	Main role	Parts you led
Hadrian Esparcia	Documenter	Part 1
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Part 1: Project Overview

1.1 The problem

Schools and organizations often lend out equipment such as projectors, extension cords, and lab tools, but they usually track it in a paper logbook or in chat messages. This makes it hard to know which items are available, who has borrowed them, and when they are due back. Items get lost or are returned late because there is no clear record. The Item Borrowing System gives the admin one place to keep the item list and the borrowing records. It lets borrowers see which items are available and send a borrow request without paperwork.

1.2 Objectives

1. To record every item and show whether it is available or borrowed.
2. To help borrowers find available items and send borrow requests without paperwork.
3. To help the admin see who borrowed each item and when it was returned.

1.3 Scope

The system WILL	The system will NOT
Keep a list of items with their name, category, and status (available or borrowed).	Handle online payments or late fees.
Record every borrow request, approval, and return.	Send email or SMS reminders.
Let borrowers search for items and see which ones are available.	Track the purchase cost or repair history of items.
Keep a borrowing history that the admin and borrowers can view.	Let borrowers reserve items for a future date.

Part 2: Users and Features

2.1 Users

User	What they do in the system
Admin	Manages the list of items, approves or rejects borrow requests, and records returns. Can also search for items and view the borrowing history.
Borrower	Searches for available items, sends borrow requests, and views their own borrowing history.

2.2 Features

No.	Feature	Used by
1	Log in to the system	Admin, Borrower
2	Search for available items	Admin, Borrower
3	Request to borrow an item	Borrower
4	View borrowing history	Admin, Borrower
5	Add a new item	Admin
6	Update or remove an item	Admin
7	Approve or reject a borrow request	Admin
8	Record the return of an item	Admin

Part 3: Use Case Diagram

3.1 The diagram

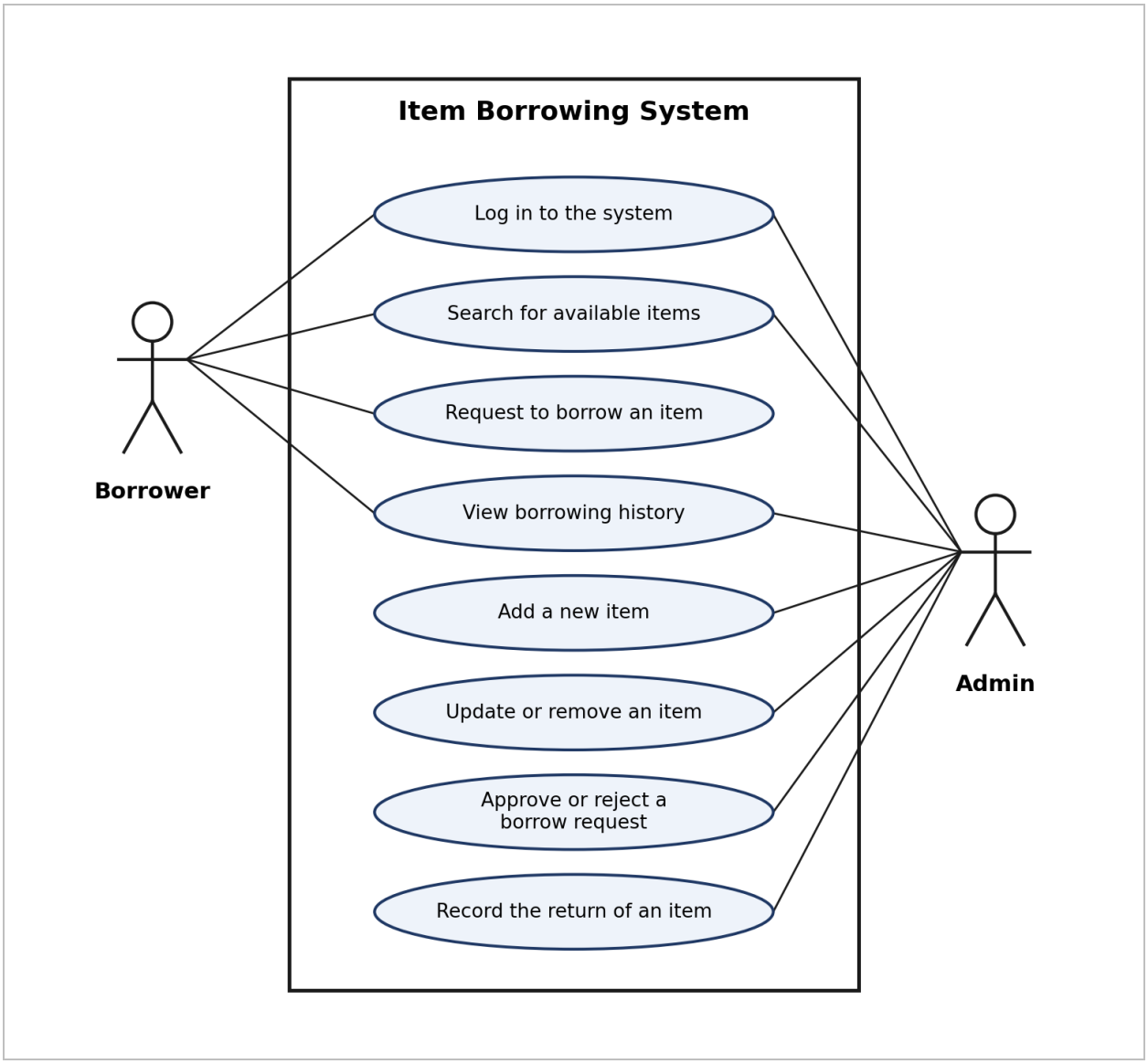


Figure 1. Use case diagram of Item Borrowing System

3.2 Use case descriptions

Use case 1

Use case name	Request to borrow an item
Actor	Borrower
Before it starts	The borrower is logged in and the item is shown as "available".
Main steps	1. The borrower searches for the item and opens its details. 2. The borrower chooses "Borrow" and enters the date the item will be returned. 3. The system checks that the item is still available. 4. The borrower confirms and the system saves the request.
Result	A new borrow request is saved with the status "pending" and the admin can see it.

Use case 2

Use case name	Approve or reject a borrow request
Actor	Admin
Before it starts	The admin is logged in and there is at least one request with the status "pending".
Main steps	1. The admin opens the list of pending requests. 2. The admin selects a request and reviews the borrower, the item, and the return date. 3. The admin chooses "Approve" or "Reject". 4. The system updates the request and, if it was approved, sets the item's status to "borrowed".
Result	The request is saved as "approved" or "rejected". An approved item is shown as "borrowed" and cannot be requested by others.

Use case 3

Use case name	Record the return of an item
Actor	Admin
Before it starts	The admin is logged in and the item has an approved borrow record that is not yet returned.
Main steps	1. The admin opens the list of borrowed items. 2. The admin selects the borrow record of the item that was brought back. 3. The admin confirms the return. 4. The system saves the return date and sets the item's status to "available".
Result	The borrow record is marked "returned" with the return date, and the item can be borrowed again.

Parts to Complete Later

These parts stay empty for now. We will fill them in during later sessions.

Part 4: Class Diagram

Your class diagram, revised from the class diagram assignment. To be completed in a later session.

Part 5: Database Design

The tables and fields your project will store. To be completed in a later session.

Part 6: Screen Design

Sketches first, then screenshots of the real screens. To be completed in a later session.

Part 7: Work Plan

Who does what, and when. To be completed in a later session.

Part 8: Testing

A few test cases and their results. To be completed in a later session.

Part 9: Final Output

User guide, link to your source code, and a short reflection. To be completed in a later session.